phu cluster and phu normalize-lineage: grouping genomes and tidying their names

A phu cluster: three vclust recipes

Viral genomes (FASTA)

all-vs-all, in one command

1 · prefilter
k-mer pairs worth aligning

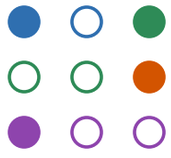
2 · align
ANI / TANI per pair (Lempel-Ziv)

3 · cluster
one algorithm per recipe

vclust [10]

--mode dereplication

cd-hit algorithm
ANI ≥ 95% · coverage ≥ 85%
prefilter identity 95%



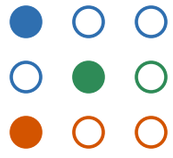
filled dot = representative

one representative per group of near-duplicates

`dereplicated-representatives.fna`

--mode votu

Leiden algorithm
ANI ≥ 95% · coverage ≥ 85%
prefilter identity 95%



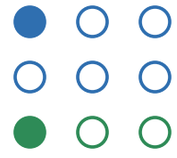
filled dot = representative

viral OTUs with MIUViG-style defaults

`clusters.tsv · cluster-representatives.fna`

--mode species

complete linkage
total ANI (TANI) ≥ 95%
prefilter identity 70%



filled dot = representative

species groups with ICTV-style defaults

`species-clusters.tsv · cluster-representatives.fna`

anything else goes through --vclust-params, straight to vclust

B phu normalize-lineage: compact codes for vConTACT predictions

A vConTACT3 assignment table keeps its columns; only the lineage cells change:

`novel_genus_1_of_Viruses`	`Viruses:NG1`
`novel_genus_1_of_novel_family_2_of_unplaced_order_within_Caudoviricetes`	`Caudoviricetes:UO:NF2:NG1`
`unplaced_phylum_within_unplaced_kingdom_within_default`	`DEF:UK:UP`

How a value is rewritten

- the anchor (the last known taxon) comes first; codes follow from the highest rank to the lowest
- N = novel node with its number · U = unplaced node (no number) · DEF = the placeholder anchor 'default'
- rank letters: R K P C O F SF G (realm to genus); each node must be of a lower rank than the one before it

Safety

- unparsed values are kept and counted in the QA summary; --strict makes them fatal
- already-compact values are left alone, so the command is idempotent
- --add-lineage appends one combined lineage column; nothing else is added or renamed

Clusters and lineage values shown are illustrative.

Recipes follow the vclust use cases; defaults follow MIUViG and ICTV practice · lineage grammar: vConTACT3 novel_/unplaced_ chains